Data Analysis Report: Attentional spatial bias in two image tasks

Junichi Nagasawa, University of Stuttgart, Kwansei Gakuin University

Sanaz Talebi, University of Stuttgart

Nikhil Pandey, University of Stuttgart

Shehab Solyman, University of Stuttgart

Table of contents

Report

Final report for eye tracking course.

Source code and data: https://github.com/jnagasawa/ET-report

Chapter 1

Background

A robust leftward bias in early visual exploration has been documented across diverse tasks, including free scene viewing, line bisection, face perception, and two-alternative forced choice paradigms (Foulsham et al. 2013; Ossandón et al. 2014). In speakers of left-to-right (LTR) languages, the proportion of initial fixations landing on the left side of a display typically falls between 60% and 93%, depending on the task and stimulus material (Ehinger et al. 2026). Because this preference operates independently of image content, it can substantially reduce the sensitivity of paradigms designed to measure content-driven attentional biases.

The origin of the leftward bias remains debated. A dominant account links it to the right-hemisphere lateralization of the attentional network, which preferentially represents the left hemifield (Corbetta and Shulman 2002). This explanation does not, however, readily account for cross-linguistic variation: speakers of Arabic and Hebrew show a weaker or reversed preference in line bisection and related tasks (Chokron and Imbert 1993; Rashidi-Ranjbar et al. 2014), suggesting that habitual reading direction modulates the baseline spatial asymmetry.

Afsari, Ossandón, and König (2016) examined this modulation directly using a priming paradigm. Bilingual participants read either RTL or LTR text passages before freely viewing photographs. Native RTL speakers (Arabic, Urdu, or Persian) showed an early rightward shift in fixation after reading RTL primes, but a leftward shift after LTR primes. Native LTR speakers who had acquired an RTL language later in life did not show comparable flexibility. These results indicate that reading direction influences spatial scanning dynamically, and that the effect depends on habit formation during native language acquisition rather than current language use alone.

Complementary evidence concerns the consequences of spatial biases for experimental design. Ehinger et al. (2026) measured first-fixation preferences in a concurrent two-stimulus paradigm with stimulus pairs placed at six angular positions relative to a central fixation cross. In the standard horizontal (9 o'clock / 3 o'clock) arrangement, German-speaking participants directed their first saccade to the left stimulus on approximately 80% of trials, with a comparable upward bias in the vertical arrangement. A diagonal placement at 1 o'clock and 7 o'clock yielded no net spatial preference, because the leftward and upward biases cancelled. Simulations based on a Bradley-Terry model showed that eliminating the spatial bias can increase statistical power by 30% or more for small attentional effects.

1.1 Motivation

The literature on spatial scanning biases has been developed almost exclusively with LTR-reading populations, and spatial biases are routinely counterbalanced rather than measured as a variable of interest. As a result, little is known about the baseline first-fixation behavior of native RTL speakers in concurrent two-stimulus paradigms. Afsari et al. (2016) showed that reading direction primes can shift the spatial bias in bilingual RTL speakers, but the baseline behavior in the absence of explicit language priming has received less attention.

The findings of Ehinger et al. (2026) raise a further question: if RTL speakers have a reduced or reversed leftward bias, their data from a standard horizontal arrangement may already be relatively unbiased, whereas LTR speakers' data from the same arrangement would suffer substantial power loss. Characterizing the bias across reading-direction groups would therefore inform stimulus placement decisions in studies that test mixed-language samples.

1.2 Research Questions and Hypotheses

We ask whether native RTL speakers (Arabic and Persian) differ from native LTR speakers in their first-fixation behavior when viewing pairs of visual stimuli in a horizontal arrangement.

We expect LTR speakers to replicate the well-documented leftward preference. For RTL speakers, we predict a reduced leftward bias, consistent with line bisection and chimeric face studies. Whether the bias reverses entirely or merely diminishes is an open empirical question.

Dependent measures: first fixation proportion (the proportion of trials on which the participant fixated each side first) and first fixation reaction time (latency from stimulus onset to the first fixation on either stimulus).

Chapter 2

Experimental Methods

2.0.1 Participants

Data collection is ongoing. Eye tracking data were recorded from 23 participants; two were excluded from all analyses after quality control identified gaze calibration problems severe enough to make their data unusable (see Quality control below). The analyzed dataset includes the remaining 21 participants: 11 who selected a left to right (LTR) language (German or English) at the start of the session, and 10 who selected a right to left (RTL) language (Arabic or Persian). Within the LTR group, 9 participants chose English and 2 chose German; within the RTL group, 7 chose Persian and 3 chose Arabic. Group membership follows this self selected session language rather than a separate native language screening; participants were recruited as native speakers of the language they chose. Age, sex, and other demographic information will be added once collection is complete.

2.0.2 Materials and stimuli

Stimuli were 16 images (000.png through 015.png), each a 1025 x 1025 px PNG, organized into eight fixed pairs. Each pair was shown six times across the session, and its position in this cycle ran independently of the six trial block structure described below, so a given pair was not tied to any particular story segment.

On each trial, the two images of a pair appeared at the same time on opposite sides of a central fixation point, along one of six axis orientations spaced 30° apart (0°, 30°, 60°, 90°, 120°, 150° from horizontal). This follows the arrangement used by Ehinger et al. (2026). Within a block of six trials, each of the six orientations occurred exactly once, in an order randomized per participant.

To keep participants engaged with both images rather than only the one they looked at first, they were asked to count the circular shapes in the pair, distinguished from a similarly sized decagon shown once during the instructions. The count could run from zero to three per pair. After each trial the experimenter entered the spoken count by key press; this count was not analyzed as a dependent measure, it served only to keep participants looking at both stimuli rather than disengaging once one had been inspected.

Alongside the eye tracking task, participants read a children's story (The Ugly Duckling) presented

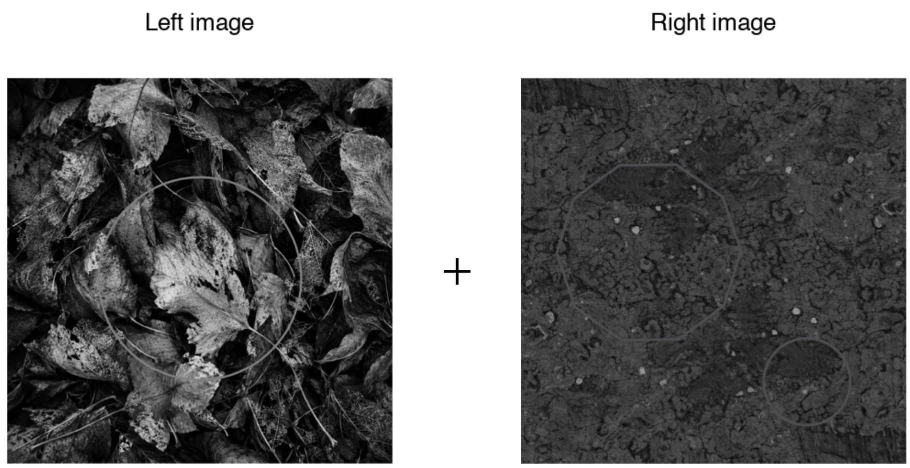

Figure 2.1: Example stimulus pair (pair 1 of 8) as presented on screen. Faint outline shapes embedded in the images are the circles participants were asked to count; the central cross marks the fixation point, not the on-screen fixation dot itself.

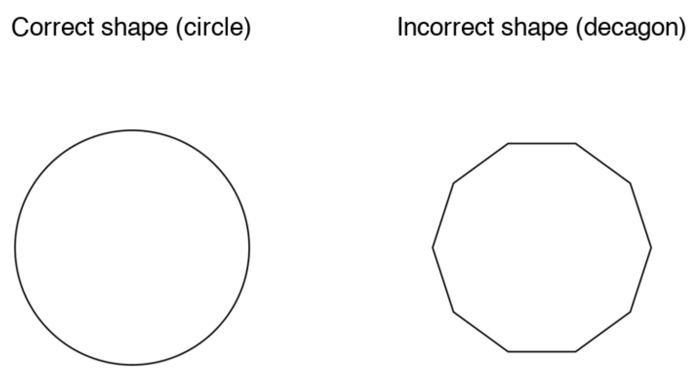

Figure 2.2: Shape distinction shown once during the instructions: the correct circle to count against the similarly sized incorrect decagon.

in nine segments. The first eight were shown one at a time, each before one six trial block; the ninth (the story's resolution) was shown once after the final trial, followed by five comprehension questions asked aloud by the experimenter and recorded manually. The story gave participants sustained exposure to text in their session language and a natural framing for the task; the comprehension questions served as an attention check rather than a scored outcome.

2.0.3 Study design

The experiment used a between subjects design with reading direction group (LTR vs RTL) as the main factor. Each participant completed 48 trials in 8 blocks of 6, with one story segment shown at the start of each block. Within a block, the six trials sampled the six axis orientations in random order, so every participant saw every arrangement exactly 8 times over the session and arrangement is fully crossed with group rather than manipulated between subjects.

2.0.4 Experiment (implementation)

The task was built in OpenSesame 4.1.14 ("Neonatal Nightingale") and run with a GazePoint GP3 HD eye tracker via PyGaze, sampling at 60 Hz (one sample every 16.7 ms). Viewing distance was fixed at 57 cm from a 1536 x 864 px display (33.8 x 27.1 cm). Before the main task, participants chose their session language from four options (Deutsch, English, Arabic, Farsi) with a number key; this choice set both the group assignment and the language of the story text and on screen instructions.

The two images making up a pair were drawn from the same 1025 x 1025 px source files, shown at a fixed 0.25 scale so that each occupied roughly 256 x 256 px on screen. Together with the placement radius used for the six arrangements, this put the two image centers about 616 px apart at every orientation, far enough that the two areas of interest described below could never overlap.

Each trial opened with a central fixation check: the script polled the tracker for up to 1.5 s for a gaze sample within 100 px of screen center, then continued regardless of whether fixation was reached. A fixed 1000 ms delay followed before the image pair appeared. Once the pair was on screen, the tracker was polled for 5000 ms; a gaze sample within 150 px of either image's center on both the horizontal and vertical axis was logged online as the first fixation, recording its side, latency, and coordinates, but the pair stayed visible for the full 5 s regardless of when a fixation was detected. Fixations were classified from the raw gaze stream by PyGaze's dispersion based algorithm, using a 1.5° dispersion threshold together with 35°/s and $9500°/s^2$ thresholds on saccade velocity and acceleration to separate a fixation from the movement bringing the eyes to it. Calibration was performed once at the start of the session; PyGaze logged accuracy (mean offset from the calibration targets, in degrees and pixels, separately for each eye) and precision (root mean square sample to sample noise) automatically for every participant.

2.1 Analysis methods

2.1.1 Preprocessing

The first fixation on each trial was identified from the raw gaze stream rather than taken from the value logged online during the session, so that the same detection rule could be applied uniformly

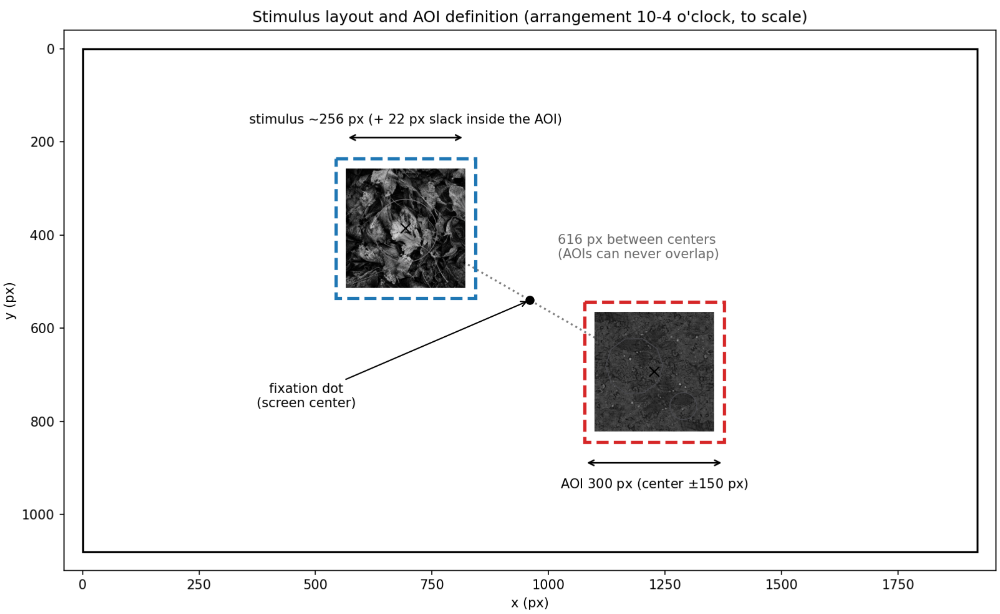

Figure 2.3: Stimulus layout and area-of-interest (AOI) definition for one arrangement (10-4 o'clock), drawn to scale. The two images are always equidistant from central fixation, along one of six axis orientations spaced 30° apart; a 150 px square margin around each image center, matching the experiment script's own online detection, defines the AOI used for first-fixation detection.

and independently of what the OpenSesame script happened to record live. GazePoint's own fixation classifier groups consecutive gaze samples into fixation events (identified by an FPOGID index, with normalized screen coordinates and a start time for each); these events were segmented into trials using the STIMULUS_ONSET and IMAGE_OFFSET markers written to the gaze log for that purpose. A fixation already in progress at stimulus onset was dropped before searching for the response, since it belongs to the preceding central fixation dot rather than to the stimuli that just appeared. Within the remaining window, the first fixation landing inside a 150 px square area of interest around either stimulus, on both axes, was taken as the trial's response, recording its side and latency; this is the same margin the experiment script used for its own online detection, applied here to tracker classified fixations rather than to individual raw samples.

Two subjects (43 and 44) were recorded before a display scaling fix was applied to the script, which left their online-logged first fixation side unusable: it registered a leftward or absent first fixation on all 96 of their combined trials, whereas the offline recomputation from the raw gaze stream found genuine rightward fixations for both of them. The offline recomputation does not depend on this part of the script and was used for every subject, not only these two, so the analysis pipeline is identical across the sample regardless of when a given participant was tested.

Trials were excluded when no fixation landed in either area of interest, or when the first fixation started less than 80 ms after onset, since a saccade that fast is planned before the stimulus appears rather than in response to it. No upper latency cutoff was applied. Slow first fixations were not random: they concentrated in arrangements without a stimulus in the upper left, so removing them would have removed a real part of the effect rather than noise; this pattern is examined on its own terms in the latency analysis below rather than treated as measurement error.

In the raw log, img_left and img_right label the stimulus on the seven-to-twelve o'clock side of an arrangement and the one on the one-to-six o'clock side, not the physical left and right of the screen; for the vertical arrangement in particular, neither stimulus sits on the horizontal meridian, so calling one of them "the right stimulus" would really mean "the bottom stimulus." To avoid this mislabeling, a trial's outcome was instead recoded as the clock hour fixated first, since the six arrangements are simply the six diameters of a clock face. Left/right and up/down bias were then derived from this hour, each skipping the one arrangement lying on the axis perpendicular to it (12 to 6 for left/right, 9 to 3 for up/down), since those two positions sit on the meridian itself and carry no information for that contrast.

2.1.2 Sensitivity check: raw samples vs. classified fixations

The preprocessing above uses `FPOGX`/`FPOGY`, GazePoint's own fixation classified gaze coordinates: consecutive raw samples are grouped into discrete fixation events (`FPOGID`) before the area of interest (AOI) test is applied. That classifier smooths and slightly delays the raw signal, so it is worth checking whether the same detection rule, applied to the unclassified gaze stream instead, changes the result. GazePoint also logs `BPOGX`/`BPOGY` ("Best Point of Gaze"), its estimate of gaze position for each raw sample, before any fixation grouping. The two are compared here directly, using the same AOI, trial window, and 80 ms anticipatory cutoff for both; the only thing that changes is whether "first fixation" means the first classified fixation event inside the AOI (FPOG) or the first raw sample inside it (BPOG):

```python
def compute_bpog_first_gaze(subj, sub_trials):
    # Same AOI test as compute_fpog_first_fixation, but walks raw
    # per-sample BPOGX/BPOGY instead of grouping into FPOGID fixations.
    tsv = pd.read_csv(f"{DATA_DIR}/{subj}.tsv", sep="\t")
    tsv.replace(["NA", "undefined"], np.nan, inplace=True)
    for c in ["TIME", "BPOGX", "BPOGY", "BPOGV"]:
        tsv[c] = pd.to_numeric(tsv[c], errors="coerce")
    tsv = tsv.dropna(subset=["BPOGX", "BPOGY"]).sort_values("TIME").reset_index(drop=True)

    onsets = tsv[tsv["USER"].astype(str).str.contains("STIMULUS_ONSET", na=False)]
    offsets = tsv[tsv["USER"].astype(str).str.contains("IMAGE_OFFSET", na=False)]["TIME"].values

    rows = []
    for (_, onset_row), (_, trial) in zip(onsets.iterrows(), sub_trials.iterrows()):
        t0 = onset_row["TIME"]
        later = offsets[offsets > t0]
        t_end = later[0] if len(later) else t0 + 10.0

        window = tsv[(tsv["TIME"] >= t0) & (tsv["TIME"] <= t_end) & (tsv["BPOGV"] == 1) &
                     tsv["BPOGX"].between(0, 1) & tsv["BPOGY"].between(0, 1)]

        lx, ly = SCREEN_W / 2 + trial["x_left"], SCREEN_H / 2 + trial["y_left"]
        rx, ry = SCREEN_W / 2 + trial["x_right"], SCREEN_H / 2 + trial["y_right"]

        side, rt_ms = "none", np.nan
        for _, s in window.iterrows():
            xi, yi = s["BPOGX"] * SCREEN_W, s["BPOGY"] * SCREEN_H
            in_left = abs(xi - lx) < AOI_MARGIN_PX and abs(yi - ly) < AOI_MARGIN_PX
            in_right = abs(xi - rx) < AOI_MARGIN_PX and abs(yi - ry) < AOI_MARGIN_PX
            if in_left or in_right:
                side = "left" if in_left else "right"
                rt_ms = (s["TIME"] - t0) * 1000.0
                break
        rows.append(dict(subject_nr=str(trial["subject_nr"]), trial_nr=int(trial["trial_nr"]),
                         bpog_first_fix_direction=side, bpog_first_fix_rt_ms=rt_ms))
    return pd.DataFrame(rows)
```

The resulting `bpog_first_fix_direction` was merged onto the same trial table, passed through the same `apply_exclusions()` used for FPOG (an 80 ms anticipatory cutoff, no upper cutoff), and compared across groups with the same hierarchical bootstrap used for the FPOG left/right and up/down proportions:

```python
trials = apply_exclusions(trials, side_col="bpog_first_fix_direction",
                          rt_col="bpog_first_fix_rt_ms",
                          min_latency_ms=MIN_LATENCY_MS,
                          reason_col="bpog_exclusion_reason", excluded_col="bpog_excluded")
```

```
df_valid_bpog = trials[~trials["bpog_excluded"]].copy()
df_valid_bpog["physical_side"] = df_valid_bpog["bpog_fixated_clock"].map(hour_side)
df_valid_bpog["physical_vert"] = df_valid_bpog["bpog_fixated_clock"].map(hour_vert)

group_lr_bpog, _ = pooled_orientation_table(df_valid_bpog, "physical_side", "right", "left")
group_ud_bpog, _ = pooled_orientation_table(df_valid_bpog, "physical_vert", "up", "down")
```

At the trial level, the two measures agreed on direction for 90.6% of trials. The disagreements were lopsided: BPOG called "right" where FPOG called "left" more than twice as often as the reverse (64 vs. 30 trials). That fits a saccade briefly overshooting or sweeping past one AOI on its way to a stable fixation elsewhere: GazePoint's fixation classifier correctly discards a sample like that, but a test that checks every raw sample cannot tell it apart from a genuine fixation. BPOG's exclusion rate under the 80 ms anticipatory cutoff was also roughly 5.6 times higher than FPOG's (7.3% vs. 1.3%), since a single raw sample can drift into an AOI for a few milliseconds in a way a fixation, which persists by definition, mostly cannot.

At the group level, the two measures agreed closely for LTR (largest difference across left/right and up/down: 0.028) and less closely for RTL (0.024 for left/right, 0.064 for up/down, where the FPOG estimate of 83.8% dropped to 77.4% under BPOG). RTL's larger movement reflects its smaller sample, 10 subjects against LTR's 11: the same handful of relabelled trials shifts a 10-subject mean more than an 11-subject one, and it falls on the same arrangements (12-6, 7-1, 8-2) that already produced the slowest, noisiest first fixations under FPOG. Still, the direction of every effect held: RTL fixated right and up more often than LTR under both measures, and RTL's upward bias, though smaller under BPOG, stayed clearly above chance. Recomputing first-fixation direction from raw samples adds noise, mostly through brief AOI crossings that are not real fixations, but it does not change any conclusion in this report. FPOG remains the measure used throughout the rest of the analysis.

2.1.3 Quality control

Before analysis, each recording was checked for complete trial markers (48 stimulus onsets, offsets, and fixation dot events), image display duration close to the intended 5 s, and effective sampling rate close to 60 Hz with few gaps longer than 50 ms. Calibration quality was estimated per trial from gaze recorded during the central fixation dot: accuracy as the mean distance from screen center, and precision as the RMS spread around each trial's own mean. These were compared across subjects with boxplots and an x/y offset scatter to catch systematic drift, and a small number of trials per subject were plotted as full scanpaths to confirm that gaze actually landed inside the areas of interest and that the arrangement positions matched the intended layout.

This check identified two participants (61 and 107, out of the 23 recorded) whose data were excluded from all analyses. Participant 61's gaze during the central fixation dot was offset from screen center by a median of 485 px, about 50% larger than any other participant's, and this offset was nearly constant across the session (interquartile range 8 px) rather than varying trial to trial as ordinary tracking noise would; scanpaths confirmed that the reconstructed gaze trace did not reach either stimulus's area of interest in any of the six arrangements, and only 8 of 48 trials registered a first fixation at all. Participant 107's calibration offset was smaller but still atypical (median 274 px, directed down and to the right of the true fixation point) and this participant was the only one in the sample whose first-fixation pattern was biased downward rather than upward. Across the other 21 participants (i.e., excluding both 61 and 107), an individual's calibration offset and their behavioral up/down bias were only weakly related (r = -0.11), so calibration drift does not account for the sample's bias in general; participant 107, however, had both the largest downward calibration offset of any participant, by more than a factor of two, and the single most extreme behavioral value in the sample, which together make a calibration artifact a more plausible explanation for this result than a genuine reversal of the group's attentional bias.

2.1.4 Analysis pipeline

Preprocessing and analysis were carried out in Python 3.11, using pandas and numpy for data handling, scipy and statsmodels for the statistical tests, and matplotlib for the figures.

Group comparisons used a hierarchical bootstrap: subjects were resampled first, then trials within each resampled subject, since trials from the same subject are not independent and treating them as such would understate the true uncertainty. The point estimate is the unweighted mean of subject means, so that a subject with more valid trials does not automatically carry more weight; the confidence interval is the 2.5th and 97.5th percentile of 5,000 such resamples, drawn with a fixed random seed so the reported intervals are exactly reproducible from the same data. This procedure was applied to the left/right and up/down fixation proportions per group. The twelve hour clock-face profile is reported as the mean of subject means without intervals, since each hour rests on roughly eight trials per subject.

A logistic regression in the style of a Bradley-Terry model estimated the log odds of fixating the clockwise-numbered stimulus first as a function of arrangement, entered as a six level categorical predictor, and reading direction group, giving a single test of whether the two groups differ once arrangement is accounted for rather than six separate group comparisons. Standard errors are clustered by subject, for the same reason the group proportions above use a hierarchical bootstrap rather than a plain proportion: trials from the same subject are not independent. Arrangement and group enter as additive main effects only: their interaction, which would give each arrangement its own group comparison, now converges (the once-empty RTL/12-6 cell now holds 3 of 78 trials), but with only 10 RTL subjects spread across twelve arrangement-by-group cells, and no clustered version of that model fit, its per-arrangement estimates are not reliable enough to report. The additive model does not have this problem and is the one reported.

First-fixation latency was examined separately by arrangement and group, since the spatial bias also showed up as a difference in response speed rather than only in which side was chosen. The latency distribution is bimodal, with a fast mode around 250 to 350 ms and a slow tail extending past 2 s; because no upper cutoff was applied during preprocessing, this tail is available for analysis rather than discarded. The median latency by arrangement and group changes little, since the fast mode dominates almost everywhere, so the two groups were also compared on the share of trials falling in the slow tail, using 2 s (the approximate valley between the two modes) as a descriptive boundary rather than an exclusion criterion.

Chapter 3

Results

The 21 participants retained after exclusions (Methods) contributed 995 valid first fixations out of 1008 trials: 11 LTR participants (521 trials) and 10 RTL participants (474 trials). The 13 dropped trials split into 9 first fixations faster than 80 ms and 4 trials with no fixation in either area of interest. All results below use the first fixation computed from FPOG, the tracker classified measure used throughout preprocessing. A sensitivity check against raw gaze samples instead of classified fixations, reported in Methods, leaves every conclusion here unchanged.

3.0.1 Left/right and up/down bias by group

Figure 3.1 and Table 3.1 give the pooled proportion of trials on which each group's first fixation went left, right, up, or down, with hierarchical bootstrap confidence intervals. `physical_side` and `physical_vert` are the same clock-hour-derived columns from Preprocessing, each skipping the one arrangement that carries no information for that axis:

```python
def pooled_orientation_table(df, axis_col, positive_label, negative_label):
    rows = []
    for group, g in df.dropna(subset=[axis_col]).groupby("group"):
        arrays = [(sg[axis_col] == positive_label).astype(float).values
                  for _, sg in g.groupby("subject_nr")]
        p_pos, lo_pos, hi_pos = hierarchical_bootstrap_ci(arrays)
        rows.append(dict(group=group, direction=positive_label, n_subjects=len(arrays),
                         n_trials=len(g), p=p_pos, ci_lo=lo_pos, ci_hi=hi_pos))
        rows.append(dict(group=group, direction=negative_label, n_subjects=len(arrays),
                         n_trials=len(g), p=1 - p_pos, ci_lo=1 - hi_pos, ci_hi=1 - lo_pos))
    return pd.DataFrame(rows)


group_lr = pooled_orientation_table(df_valid_fpog, "physical_side", "right", "left")
group_ud = pooled_orientation_table(df_valid_fpog, "physical_vert", "up", "down")
group_dir = pd.concat([group_lr, group_ud], ignore_index=True)
```

Table 3.1: Pooled first-fixation direction by group (FPOG-based), with hierarchical bootstrap 95% confidence intervals.

Group	Direction	P(first fixation)	95% CI
LTR	left	71.0%	[61.5%, 79.7%]
LTR	right	29.0%	[20.3%, 38.5%]
LTR	up	71.6%	[61.4%, 81.0%]
LTR	down	28.4%	[19.0%, 38.6%]
RTL	left	50.9%	[38.7%, 62.5%]
RTL	right	49.1%	[37.5%, 61.3%]
RTL	up	83.8%	[77.2%, 89.3%]
RTL	down	16.2%	[10.7%, 22.8%]

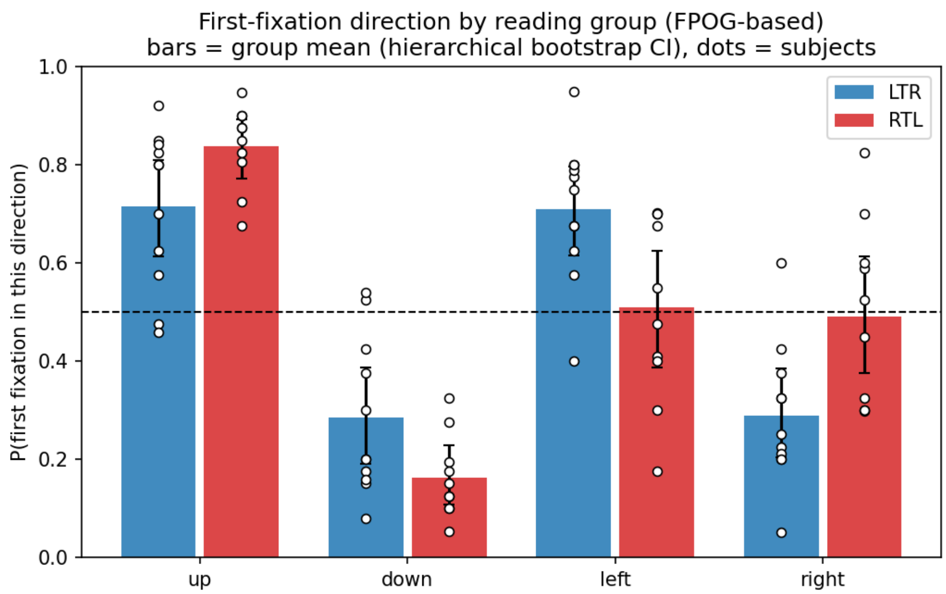

Figure 3.1: First-fixation direction by reading group. Bars are the group mean (hierarchical bootstrap CI); dots are individual subject means; the dashed line marks chance.

Both groups fixate up well above chance. On left/right, LTR does too, but RTL's raw proportion no longer distinguishes it from chance: its CI [38.7%, 62.5%] spans 50%, and its point estimate (50.9%) sits almost exactly on it. LTR's leftward bias (71.0%) sits inside the 60% to 93% range reported for comparable paradigms in the literature reviewed in the Introduction; RTL's (50.9%) is smaller and in the predicted direction, but barely so, and the two groups' CIs overlap by a wide margin, so this contrast alone cannot distinguish a real group difference from noise. The up/down contrast runs the other way: RTL's upward bias (83.8%) is if anything larger than LTR's (71.6%), which the Introduction's hypotheses did not specifically anticipate.

3.0.2 Clock-face profile

Figure 3.2 breaks the same bias down by all twelve clock positions rather than collapsing to left/right and up/down. Each trial contributes to the two hours actually presented in its arrangement, so every hour's estimate is conditioned on having been on screen at all:

```
rows = []
for _, r in df_valid_fpog.dropna(subset=["fixated_clock"]).iterrows():
    for h in r["arrangement"]:
        rows.append(dict(group=r["group"], subject_nr=r["subject_nr"], hour=int(h),
                          fixated=float(r["fixated_clock"] == h)))
clock_long = pd.DataFrame(rows)

subj_clock = (clock_long.groupby(["group", "subject_nr", "hour"])["fixated"]
              .agg(["mean", "count"]).reset_index())
group_clock = (subj_clock.groupby(["group", "hour"])["mean"]
               .mean().reset_index(name="p"))  # mean of subject means
```

Both groups trace a similar shape: fixation probability is highest for the top three to four hours (10 through 1) and lowest for the bottom two (5 through 7), consistent with the up/left preference already visible in the pooled proportions.

A few individual hours diverge more than the pooled contrasts suggest, although each hour rests on about eight trials per subject and is correspondingly noisier. At hour 2 (upper right), RTL fixates first on 78.2% of trials against LTR's 36.4%, RTL's largest single-hour lead over LTR anywhere on the clock. At hour 8 (lower left), the pattern reverses, by almost exactly the same margin: LTR leads 63.6% to 21.8%. Hour 6 (bottom, on the vertical meridian) is the most extreme point on the plot: LTR still fixates it first on 13.6% of trials, against just 3 of 78 for RTL.

3.0.3 Individual-subject bias vectors

Figure 3.3 plots each subject as a single point: horizontal bias (P(right) minus P(left)) against vertical bias (P(up) minus P(down)), each rescaled from a 0 to 1 proportion to a -1 to 1 bias so that 0 means no preference:

```
subj_rows = []
for (group, subj), g in df_valid_fpog.groupby(["group", "subject_nr"]):
    px = (g["physical_side"].dropna() == "right").astype(float).values
    py = (g["physical_vert"].dropna() == "up").astype(float).values
    subj_rows.append(dict(group=group, subject_nr=subj,
                           x=2 * px.mean() - 1, y=2 * py.mean() - 1))
quad_subj = pd.DataFrame(subj_rows)

group_rows = []
for group, g in df_valid_fpog.groupby("group"):
    side_arrays = [(sg["physical_side"].dropna() == "right").astype(float).values
                   for _, sg in g.groupby("subject_nr")]
    vert_arrays = [(sg["physical_vert"].dropna() == "up").astype(float).values
```

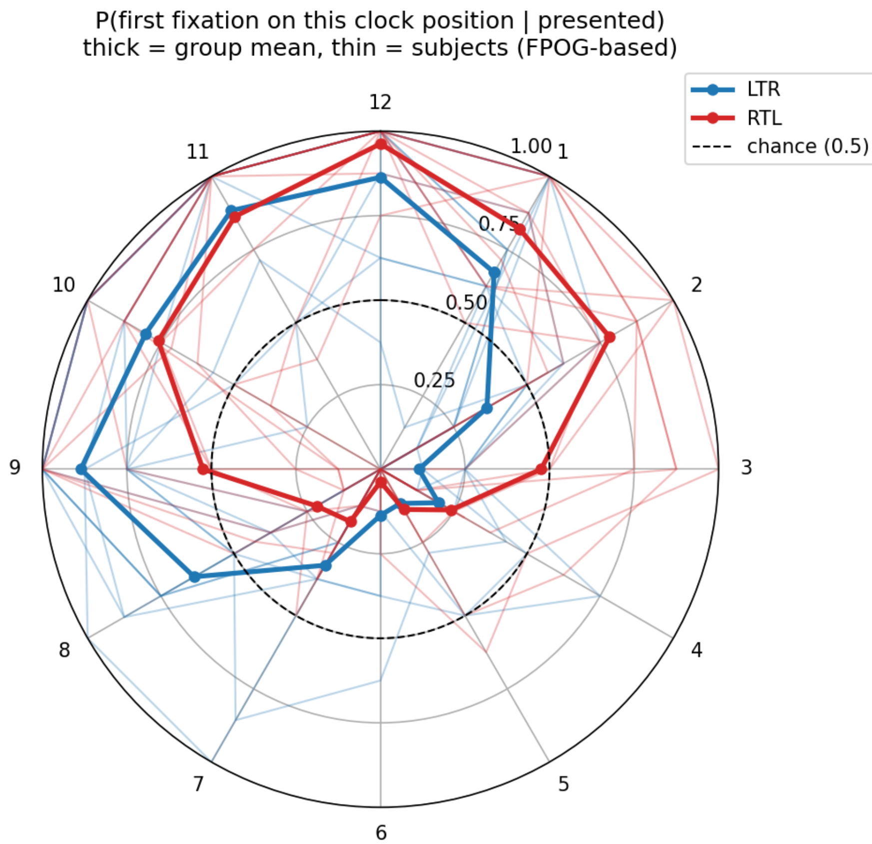

Figure 3.2: P(first fixation on this clock position, given it was presented), by group. Thick lines are group means; thin lines are individual subjects.

```
                    for _, sg in g.groupby("subject_nr")]
    px, px_lo, px_hi = hierarchical_bootstrap_ci(side_arrays)
    py, py_lo, py_hi = hierarchical_bootstrap_ci(vert_arrays)
    group_rows.append(dict(group=group, x=2 * px - 1, y=2 * py - 1,
                            x_ci_lo=2 * px_lo - 1, x_ci_hi=2 * px_hi - 1,
                            y_ci_lo=2 * py_lo - 1, y_ci_hi=2 * py_hi - 1))
quad_group = pd.DataFrame(group_rows)
```

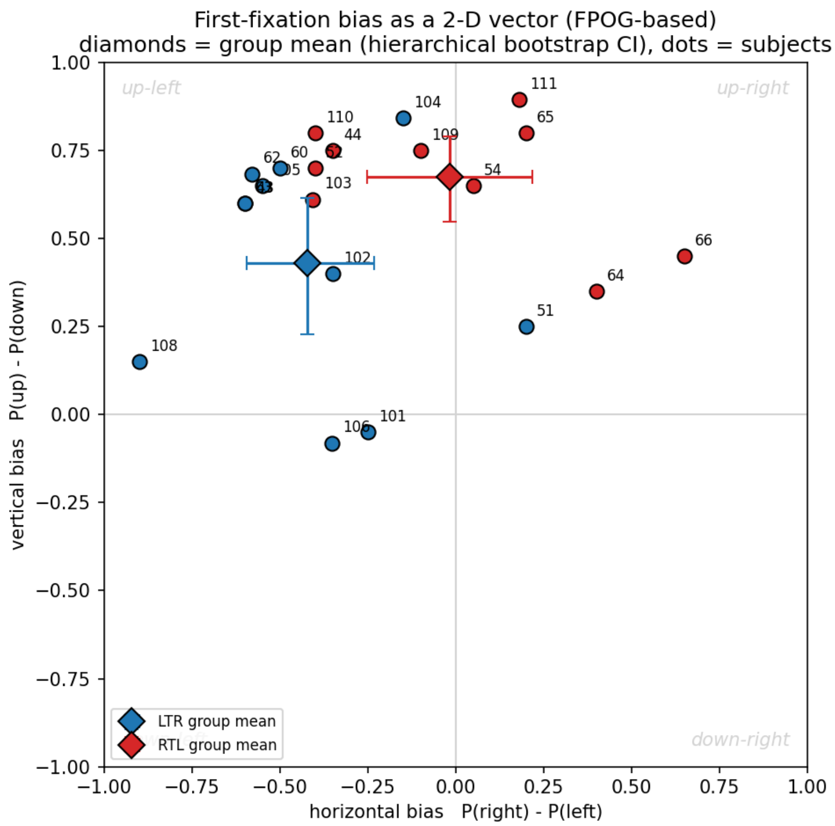

Figure 3.3: First-fixation bias as a 2-D vector. Diamonds are group means with hierarchical bootstrap CIs; dots are individual subjects.

Both group means fall in the up-left quadrant, though RTL's now sits essentially on the vertical axis itself: its mean horizontal bias is -0.02. LTR's confidence intervals in both dimensions exclude zero, so LTR is reliably biased up and left as a group. RTL's vertical CI also excludes zero, but its horizontal CI [-0.25, 0.22] spans zero, so the group-level leftward lean cannot be distinguished from no preference on this measure alone, matching Table 3.1. At the individual level, six subjects have a positive horizontal bias, a net rightward preference: RTL subject 54 (x = 0.05, the smallest lean of either sign in the sample), RTL subject 111 (x = 0.18), RTL subject 65 and LTR subject 51 (both x

= 0.20), RTL subject 64 (x = 0.40), and RTL subject 66 (x = 0.65), the largest rightward lean of any subject in either group. Two LTR subjects (101, 106) sit just below zero on the vertical axis (-0.05 and -0.08), the only two participants without an upward preference; every RTL subject shows a positive vertical bias (0.35 to 0.89).

3.0.4 A single test of the group difference

The proportions above answer four separate questions (left/right and up/down, for each group) rather than the one question of primary interest: do LTR and RTL differ once each arrangement's own baseline preference is accounted for? Methods describes a logistic regression built for exactly this: the log odds of fixating the clockwise-numbered stimulus in each pair (`img_right`, the one-to-six o'clock member), as a function of arrangement (six levels) and group:

```
import statsmodels.formula.api as smf

df_bt = df_valid_fpog.copy()
df_bt["y_right"] = (df_bt["fpog_first_fix_direction"] == "right").astype(int)
df_bt["arrangement_str"] = df_bt["arrangement"].astype(str)

bt_model = smf.logit("y_right ~ C(arrangement_str) + C(group)", data=df_bt).fit(
    disp=0, cov_type="cluster", cov_kwds={"groups": df_bt["subject_nr"]})
```

The model converges cleanly ($n = 995$); standard errors are clustered by subject, since trials from the same subject are not independent – the same reason the proportions above use a hierarchical bootstrap rather than a plain proportion. Arrangement drives most of the fit: relative to the baseline arrangement, the two arrangements without an upper-left stimulus (7 and 1 o'clock; 8 and 2 o'clock) sharply raise the odds of fixating the clockwise-numbered member of the pair ($b = 2.46$, $p < .001$; $b = 1.62$, $p < .001$), two of the three arrangements with the largest slow latency tails below. The remaining two non-baseline arrangements (11 and 5 o'clock; 12 and 6 o'clock) lower it instead ($b = -0.67$, $p = .005$; $b = -1.04$, $p = .018$); only 9 and 3 o'clock does not differ from baseline.

Group now reaches conventional significance: the RTL coefficient is $b = 0.93$ (cluster SE = 0.42, $p = .027$). Fit without clustering, the same coefficient comes out $p < .001$, which would overstate the group difference; that version treats all 995 trials as independent observations, which overstates precision given only 21 subjects, so it is not the one reported. The clustered result agrees with the descriptive picture in Table 3.1 – a difference in the predicted direction – but goes further: once each arrangement's own baseline pull is accounted for, the group difference clears the .05 threshold, even though the raw left/right proportions alone (RTL barely off 50%) would not have suggested it.

An arrangement by group interaction, which would give each arrangement its own group comparison instead of one pooled test, was checked and not used:

```
inter_model = smf.logit("y_right ~ C(arrangement_str) * C(group)", data=df_bt).fit(disp=0)
# converges now that every arrangement x group cell has at least one trial
# (the (12, 6)/RTL cell, previously empty, now holds 3 of 78)
```

Every cell is now populated, so the interaction model converges. It still isn't the reported result: with only 10 RTL subjects spread across twelve arrangement-by-group cells, and no clustered version of this model fit, its per-arrangement estimates aren't reliable enough to report alongside the clustered additive model above, which is what's used instead.

3.0.5 First-fixation latency by arrangement

Figure 3.4 shows the share of trials with a first fixation slower than 2000 ms, separately by arrangement and group; 2000 ms is a descriptive boundary at the valley of the bimodal latency distribution, not an exclusion criterion:

```
SLOW_MS = 2000

lat = df_valid_fpog.dropna(subset=["fpog_first_fix_rt_ms"]).copy()
lat["arrangement_lbl"] = lat["arrangement"].apply(lambda a: f"{a[0]}-{a[1]}")
lat["slow"] = lat["fpog_first_fix_rt_ms"] > SLOW_MS

med = lat.groupby(["group", "arrangement_lbl"])["fpog_first_fix_rt_ms"].median().unstack(0)
pslow = lat.groupby(["group", "arrangement_lbl"])["slow"].mean().unstack(0)
```

Median latency for both groups (`med` above) stays within 248 to 288 ms across every arrangement and is not shown here, since the fast mode dominates almost everywhere and the group difference lives in the slow tail (`pslow`) instead.

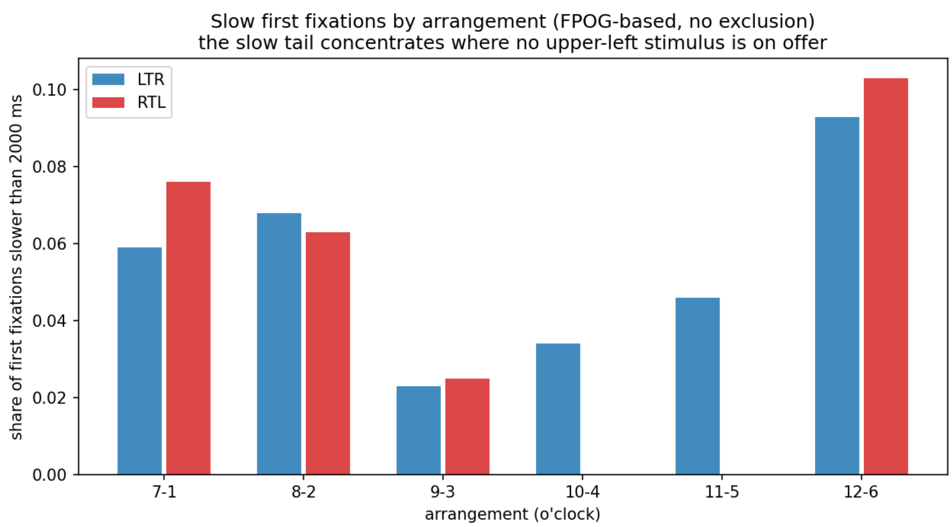

Figure 3.4: Share of first fixations slower than 2000 ms, by arrangement and group.

The slow tail concentrates in the three arrangements without an upper-left stimulus (7-1, 8-2, 12-6) for both groups. RTL's slow tail is somewhat larger than LTR's in two of the three: 7.6% versus 5.9% for 7-1, its largest gap anywhere, and 10.3% versus 9.3% for 12-6; the third, 8-2, now runs slightly the other way, with RTL's tail (6.3%) a touch below LTR's (6.8%). In the three arrangements that do offer an upper-left stimulus (9-3, 10-4, 11-5), both groups are fast and the slow tail nearly disappears, reaching exactly 0% for RTL in 10-4 and 11-5. Where the choice proportions above show only a direction consistent with a group difference, latency points the same way for 7-1 and 12-6: RTL takes longer than LTR specifically where its preferred upper-left target is unavailable, though the effect is not uniform across all three such arrangements – 8-2 in fact runs slightly the other way. These are raw shares with no confidence intervals, and each RTL cell holds 78 to 80 trials, so they carry the same caveat about sample size as the proportions do.

Chapter 4

Discussion

LTR speakers replicated the well-documented leftward bias, and both groups fixated upward well above chance. RTL speakers' raw leftward proportion sat almost exactly at chance and, on its own, was not distinguishable from no preference at all. The single confirmatory test, a logistic regression with clustered standard errors, told a different story once each arrangement's own baseline pull was accounted for: group reached conventional significance ($b = 0.93$, $p = .027$). Arrangement itself still explained far more of the variance than group did, and three largely independent measures agreed on direction even where they disagreed on strength: the choice proportions, the individual bias vectors, and the latency data all pointed the same way, toward a reading-direction effect that the confirmatory test now supports directly rather than only suggesting.

4.0.1 Relation to the literature

LTR's leftward proportion (71.0%) sits inside the 60-93% range reported by Foulsham et al. (2013) and Ossandón et al. (2014), though below the roughly 80% Ehinger et al. (2026) found for German speakers in the same horizontal arrangement. Part of that gap is probably composition: our LTR group is nine English and two German speakers, not a purely German sample, and the counting task asked participants to engage with both images rather than just glance and move on. Either could pull the estimate down without contradicting the earlier result.

RTL's proportion (50.9%) fits a "reduced, not reversed" account: Chokron and Imbert (1993) and Rashidi-Ranjbar et al. (2014) found weaker or reversed leftward bias in Arabic, Hebrew, and mixed-reading-habit populations doing line bisection, and ours lands almost exactly at chance rather than reversing outright. Afsari et al. (2016) showed native RTL readers shift rightward after an RTL priming passage but leftward after an LTR one, while late-acquired RTL readers show no such flexibility. Our RTL participants were unprimed, so a near-chance default fits the baseline that priming would move away from in either direction. That the raw proportion sits at chance while the arrangement-adjusted regression still finds a reliable group difference ($b = 0.93$, $p = .027$) suggests the horizontal habit is there but weak enough to be swamped, in the simple proportion, by each arrangement's own pull.

The vertical result is the one the hypotheses did not anticipate. RTL's upward bias (83.8%) is, if anything, larger than LTR's (71.6%), and reading direction is a horizontal habit; neither the right-hemisphere lateralization account (Corbetta and Shulman 2002) nor the cross-linguistic account predicts a group difference on this axis. The arrangement effects in the logistic regression suggest a possible reason. Two of the three arrangements without an upper-left stimulus, 7-1 and 12-6, are where RTL's slow-latency tail balloons and where its choice proportions diverge most from LTR's – most strikingly at 12-6, where RTL fixates the lower stimulus first on just 3 of 78 trials. That pattern reads as a fallback: when an arrangement offers no upper-left target for the horizontal bias to lock onto, both groups default more heavily to the vertical one, and RTL leans on it hardest because it has less horizontal pull to begin with. The data are consistent with this account but do not confirm it: the arrangement-by-group interaction that would test it directly now converges, but with only 10 RTL subjects spread across twelve cells and no clustered fit computed, it isn't a reliable per-arrangement test (see Limitations). If correct, it would mean the upward bias is closer to a general property of the attentional system than a culturally modulated one, a distinction the horizontal-axis literature reviewed in the Introduction does not address.

Individual variability sits underneath all of this. Foulsham et al. (2018) report that leftward viewing biases are stable within a person but unrelated to handedness or line-bisection performance, i.e., a real trait that doesn't reduce cleanly to the usual explanations. Our quadrant plot shows the same texture: six subjects with a net rightward lean regardless of group, and two LTR subjects with no upward preference at all, the only two in the sample. The group means are real, but they sit on top of individual variation the group-level story doesn't reach.

4.0.2 Limitations

The RTL group is still smaller than LTR's, ten subjects and 474 trials against eleven and 521, and every RTL confidence interval reflects it: the descriptive left/right proportion, for instance, is close enough to chance that it cannot be told apart from no preference on its own. Clustering standard errors by subject, the correct choice given non-independent trials, is still what separates the logistic regression's naive $p < .001$ from its properly clustered $p = .027$: the group effect survives clustering this time, but the margin over the .05 threshold is not large, and a smaller or differently composed sample could still have landed on the other side of it. It is the same power problem Ehinger et al. (2026) describe for spatial-bias designs generally, here limiting how confidently the group comparison can be interpreted even where it does reach significance.

Group membership follows the language a participant selected at the start of the session, not a verified native-language history or a handedness screen, so a lateralization confound cannot be ruled out directly. Reading direction is also confounded with everything else that differs between four languages and their speakers: culture, script, the translated story text, the recruitment pool. A between-subjects design with 21 people cannot separate reading direction from these co-varying factors; it can only describe the group difference, not attribute it.

The arrangement-by-group interaction, which would give each arrangement its own group comparison and test the "vertical fallback" account directly, now converges – the previously empty 12-6/RTL cell now holds 3 of 78 trials – but with only 10 RTL subjects spread across twelve cells, and no clustered version of the model fit, it isn't a reliable per-arrangement test. The hour-by-hour and arrangement-by-arrangement differences described above remain descriptive, not confirmed by a formal test.

The counting task required participants to engage with both images rather than only the one fixated first, which is a deliberate departure from pure free viewing. First-fixation latency stayed fast (roughly 248-288 ms median in every arrangement), which argues against much top-down influence on the very first saccade, but it does not rule out a subtler effect of knowing a count would be required. Participant 107's atypical calibration offset, discussed in Methods, leaves a small residual uncertainty around that participant's contribution to the vertical-bias estimates. Finally, demographic variables like age, sex, and handedness are not yet available for this sample.

4.0.3 Challenges during the project

The clock-face recoding turned out to be necessary, not optional. The raw log's `img_left`/`img_right` labels track each pair's position along the clock diameter, not physical screen sides, so for every non-horizontal arrangement one of "left" and "right" was really "top" or "bottom." This surfaced only once the vertical and diagonal arrangements were analyzed, and the fix, recoding every trial by the clock hour actually fixated, had to happen before any left/right or up/down contrast could be trusted.

Two participants (43, 44) were run before a display-scaling bug was caught, and it made their online-logged first-fixation values unusable: both showed a leftward or absent first fixation on all 96 combined trials, which offline recomputation from the raw gaze stream showed to be wrong in both cases. That recomputation became the pipeline used for every participant, not just these two, once the online log could no longer be trusted uniformly.

Recruiting native RTL speakers was harder than recruiting LTR speakers, which is the direct cause of the 11-to-10 imbalance and, downstream, of the power problem described above. Quality control also caught calibration problems in two participants (61, 107) only after full data collection, when the sessions could no longer be corrected. And running the study in four languages (German, English, Arabic, Persian) meant translating the story text, instructions, and comprehension questions into all four, coordinated across a small team, on top of the core eye-tracking design.

4.0.4 Outlook

The most direct next step is recruiting one more RTL participant to reach parity with the LTR group. That alone would narrow the RTL confidence intervals further and give the arrangement-by-group interaction enough subjects per cell to support a clustered fit. The group effect in the logistic regression ($b = 0.93$, $p = .027$) already clears conventional significance in the current sample; parity would make that result more robust rather than establish it for the first time. Adding a handedness measure and a verified native-language history, rather than relying on session-language selection, would rule out the most obvious lateralization confound directly instead of leaving it as a design limitation.

The vertical-fallback account is testable on its own: a follow-up powered specifically for an arrangement (upper-left target present vs. absent) by group interaction would confirm or rule it out, rather than relying on the descriptive pattern reported here. It also suggests a concrete revision to Ehinger et al.'s (2026) stimulus-placement recommendation for mixed-language samples: rather than picking one diagonal arrangement that is neutral for a single population, a design could place each group's preferred target so both groups' biases are equally engaged, since the arrangements that produced the largest group divergence here were exactly the ones offering neither group its usual upper-left anchor.

Two things would be worth changing about the pipeline itself. Logging clock-hour position directly, instead of `img_left`/`img_right`, would remove the recoding step and the ambiguity that could have gone unnoticed if the vertical and diagonal arrangements had not been checked closely. And running the offline recomputation against a few pilot sessions before full data collection would have caught the display-scaling bug immediately, rather than after two participants' data needed a separate fix.

Chapter 5

Contribution table

Task	Junichi	Sanaz	Nikhil	Shehab
Background Literature	-	-	—	—
Experiment Design	-	-	—	—
Stimulus Design	—	—	-	-
Piloting	—	—	—	-
Data-Recording	—	-	-	—
Non-Final-Talk presenting (who talks)	—	-	—	-
Non-Final-Talk presenting (who prepares)	—	-	—	-
Final-Talk presenting (who talks)	—	—	—	-
Final-Talk presenting (who prepares)	—	-	-	-
Data Analysis Scripts	—	-	—	-
Report Writing	—	—	-	-
Report Lectorate	—	-	-	-

References

Afsari, Zaeinab, José P. Ossandón, and Peter König. 2016. "The Dynamic Effect of Reading Direction Habit on Spatial Asymmetry of Image Perception." *Journal of Vision* 16 (11): 8. https://doi.org/10.1167/16.11.8.

Chokron, Sylvie, and Michel Imbert. 1993. "Influence of Reading Habits on Line Bisection." *Cognitive Brain Research* 1 (4): 219–22. https://doi.org/10.1016/0926-6410(93)90005-P.

Corbetta, Maurizio, and Gordon L. Shulman. 2002. "Control of Goal-Directed and Stimulus-Driven Attention in the Brain." *Nature Reviews Neuroscience* 3 (3): 201–15. https://doi.org/10.1038/nrn755.

Ehinger, Benedikt, Maximilian Dominik Bernhardt, Simon Enkel, et al. 2026. *A Simple Spatial Bias Modification to Increase Statistical Power in Concurrent Two Alternatives Choice Tasks*.

Foulsham, Tom, Emma Frost, and Laura Sage. 2018. "Stable Individual Differences Predict Eye Movements to the Left, but Not Handedness or Line Bisection." *Vision Research* 144: 38–46. https://doi.org/10.1016/j.visres.2018.02.002.

Foulsham, Tom, Alasdair Gray, Eleni Nasiopoulos, and Alan Kingstone. 2013. "Leftward Biases in Picture Scanning and Line Bisection: A Gaze-Contingent Window Study." *Vision Research* 78: 14–25. https://doi.org/10.1016/j.visres.2012.12.001.

Ossandón, José P., Selim Onat, and Peter König. 2014. "Spatial Biases in Viewing Behavior." *Journal of Vision* 14 (2): 20. https://doi.org/10.1167/14.2.20.

Rashidi-Ranjbar, Neda, Mahdi Goudarzvand, Shahram Jahangiri, Peter Brugger, and Tobias Loetscher. 2014. "No Horizontal Numerical Mapping in a Culture with Mixed-Reading Habits." *Frontiers in Human Neuroscience* 8: 72. https://doi.org/10.3389/fnhum.2014.00072.